Data understanding

Data source and relational integration

ENDES 2007–2025 (INEI): modules RECH0, RECH1, RECH6, RECH23 and REC41 (via REC21), joined by HHID / CASEID → N = 304,271 children under 5 years of age

Target variable and data leakage prevention

Y = 1 if HAZ < −2 SD (WHO); prevalence 15.8% (1 : 5.3). Height (HC3), weight (HC2) and variables recorded after the measurement are excluded

Data preparation

Preprocessing

Harmonization of codes across years · missing values · categorical encoding · child–mother consistency > 99.9%

Four-stage feature selection

189 columns → S1: missing > 60% · S2: |r| > 0.80 and VIF > 5 · S3: χ^2 and Kruskal–Wallis ($p < 0.05$) → 74 → S4: gradient boosting importance → **45 variables** (+ sufficiency check)

Modeling

Class imbalance handling

Class weights (balanced / scale_pos_weight) · INEI expansion factor (HV005) for representativeness · triage-oriented decision threshold τ

Model training and comparison

Logistic regression (L2), CART, XGBoost, CatBoost, MLP 3×200 and LightGBM · stratified 5-fold cross-validation

Evaluation

Performance evaluation

AUC-ROC, AUC-PR, sensitivity, precision, specificity and F1 · 95% bootstrap CI (1,000) · DeLong test

SHAP explainability

TreeExplainer · local explanation and aggregated by department and natural region

Deployment

Implementation and reproducibility

Streamlit web application (3 layers) · Docker, Makefile and Zenodo

Relevance assessment

Panel of 4 health professionals · TAM2-based instrument